Young-Tak Kim, PhD

Clinical AI Safety | Human–AI Decision-Making | Biomedical Informatics

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Research Profile

Biomedical AI researcher focused on the safe and effective implementation of AI in clinical care while preserving clinicians' expertise and independent judgment. Research spans medical imaging, generative and multimodal AI, biomedical knowledge graphs, and clinical education, combining computational development with clinician-centered evaluation. Current work examines operational safety, uncertainty communication, and clinician–AI interaction. The goal is to develop and prospectively evaluate AI tools within clinical workflows, assessing their clinical usefulness and unintended consequences, including effects on clinicians' decision-making, learning, and reliance on AI.

Research Appointments

Research Fellow, Radiology — *Boston, MA*

Massachusetts General Hospital / Harvard Medical School — *Sep 2024 – present*

- Laboratory of Medical Imaging and Computation
  Mentor: Synho Do.

Research Professor, Biomedical Informatics — *South Korea*

Korea University College of Medicine — *Sep 2022 – Jul 2024*

- Cancer Genomics Laboratory
  Mentor: Jason K. Sa.

Education

PhD in Brain and Cognitive Engineering — *Seoul, South Korea*

Korea University — *Aug 2022*

- Thesis: Minimally-Invasive Neuromonitoring Enhanced by Artificial Intelligence: Neurosurgical Applications in Traumatic Brain Injury.

BEng in Computer and Communication Engineering — *Seoul, South Korea*

Korea University — *Feb 2014*

Research Funding

Development of a real-time generative artificial intelligence in medical imaging based on physiological signals: a novel paradigm application of neuroimaging monitoring for severe brain and neurological disorders — *National Research Foundation of Korea (NRF)*

Sejong Science Fellowship — *Mar 2023 – Jul 2025*

- Role: Principal Investigator

Teaching Experience

Introduction to Artificial Intelligence (AAI101) — *Seoul, South Korea*

Graduate School of SW·AI Convergence, Korea University — *Fall 2022; Spring 2023*

- Course instructor

Journal Articles

The applicant's name is **bold and underlined**; * denotes equal authorship contribution.

When AI says "I don't know": a multireader multicase study of safety-zone guidance and radiologists' reliance on AI advice in screening mammography *2026*

Kim YT, Kim H, Gee MS, Bahl M, Do S.

- *BMJ Quality & Safety* **(Under review)**.

Lesion-Specific Artificial Intelligence Scores on Screening Digital Breast Tomosynthesis Are Associated with Breast Cancer Aggressiveness *2026*

Kim YT, Kim H, Dwan D, Do S, Bahl M.

- *Radiology* **(Under review)**.

Literature-to-metagenome alignment for microbiome discovery across human diseases *2026*

Kim YT, Kim H, Do S.

- *Microbiological Research* **(Under review)**.

EEG network features and APOE ε4 for amyloid classification in mild cognitive impairment *2026*

Kong J, So M, Lee I, Park H, **Kim YT**, Kim HJ, Kim H, Kang SH, Do S, Lim DK, Lee CN, Kim JB.

- *Scientific Reports*. Published online November 4, 2026.

Lateralized amyloid PET uptake and beta-band connectivity in relation to cognition in temporal lobe epilepsy *2026*

Kong J*, Kim HJ*, So M, Lee I, Kim H, **Kim YT**, Kim HJ, Do S, Lim DK, Han JH, Kang SH, Pahk K, Kim JB.

- *NeuroImage*. 2026;341:122286.

League of Radiologists—an End-to-End AI Framework for Scalable and Gamified Radiology Education: A Pilot Implementation in Chest Radiography *2026*

Kim H*, **Kim YT***, Langarica S*, Fialkowski KP, Seah JCY, Tang JSN, Song KD, Jung DC, Bae KT, Cochran RL, Succi MD, McDermott S, Bahl M, Ackman JB, Lev MH, Gee MS, Do S.

- *Journal of Imaging Informatics in Medicine*. Published online April 20, 2026.

The Effects of Prompt Engineering on a Multimodal Large Language Model in Neuroradiology: A Case Study of GPT-4o for Intracranial Hemorrhage Diagnosis from Brain CT *2026*

Kim YT*, Kim H*, Kim H, Lee S, Lim DK, Han JH, Kim JB, Do S.

- *KSII Transactions on Internet and Information Systems*. 2026;20:1966–1983.

Targeted Surface-Enhanced Raman Scattering for Highly Accurate Identification of Bacterial Species and Finding Spectral Signatures with Explainable Artificial Intelligence *2026*

Kim YT, Cho JE, Hwang MJ, Kaushal S, Singhal N, Kim JB, Do S, Lim DK.

- *ACS Nano*. 2026;20:13636–13648.

Defining operational safety in clinical artificial intelligence systems *2026*

Kim YT, Kim H, Bahl M, Lev MH, González RG, Gee MS, Do S.

- *npj Digital Medicine*. 2026;9:281.

A Deep-Learning Framework for Predicting Teprotumumab Treatment Response in Thyroid Eye Disease *2026*

Langarica S, Lee NG, Alkhadrawi AM, **Kim YT**, Ha SK, Do S, Lin LY.

- *Ophthalmology Science*. 2026;6:101098.

Advanced side-channel evaluation using contextual deep learning-based leakage modeling *2026*

Alabdulwahab S, Kim JC, **Kim YT**, Son Y.

- *ACM Transactions on Software Engineering and Methodology*. 2026;35:1–20.

ARANet: Adaptive resolution attention network for precise MRI-based segmentation and quantification of fetal size and amniotic fluid volume *2026*

Alkhadrawi AM, Peña-Trujillo V, Gallo-Bernal S, Gee MS, Jaimes Cobos CE, Kim K, Langarica S, **Kim YT**, Victoria T, Do S.

- *Journal of Imaging Informatics in Medicine*. 2026;39:1202–1216.

Response to 'Methodological and statistical concerns in MINERVA microbiome–disease knowledge graph' by Salvatore Chirumbolo *2025*

Langarica S, **Kim YT**, Alkhadrawi A, Kim JB, Do S.

- *Briefings in Bioinformatics*. 2025;26:bbaf671.

Electroencephalography functional network for screening amyloid positivity in mild cognitive impairment: A cross-sectional study *2025*

Kong J, So M, Park H, **Kim YT**, Kim H, Pahk K, Kang SH, Do S, Lim DK, Lee CN, Kim JB.

- *Clinical Neurophysiology*. 2025;180:2111374.

Non-contrast CT-based pulmonary embolism detection using GAN-generated synthetic contrast enhancement: Development and validation of an AI framework *2025*

Kim YT, Bak SH, Han SS, Son Y, Park J.

- *Computers in Biology and Medicine*. 2025;198:111109.

MINERVA—microbiome network research and visualization atlas: A scalable knowledge graph for mapping microbiome–disease associations *2025*

Langarica S, **Kim YT**, Alkhadrawi A, Kim JB, Do S.

- *Briefings in Bioinformatics*. 2025;26:bbaf472.

Enhancing deep learning-based side-channel analysis using feature engineering in a fully simulated IoT system *2025*

Alabdulwahab S, Cheong M, Seo A, **Kim YT***, Son Y*.

- *Expert Systems with Applications*. 2025;266:126079.

HKA-Net: Clinically adapted deep learning for automated measurement of hip–knee–ankle angle on lower limb radiography for knee osteoarthritis assessment *2024*

Kim YT*, Han BS*, Kim JB, Sa JK, Hong JH, Son Y, Han JH, Do S, Chae JS, Bae JK.

- *Journal of Orthopaedic Surgery and Research*. 2024;19:777.

Privacy-preserving synthetic data generation method for IoT-sensor network IDS using CTGAN *2024*

Alabdulwahab S, **Kim YT**, Son Y.

- *Sensors*. 2024;24:7389.

Increased cerebellar vermis volume following repetitive transcranial magnetic stimulation in drug-resistant epilepsy: A voxel-based morphometry study *2024*

So M, Kong J, **Kim YT**, Kim KT, Kim H, Kim JB.

- *Frontiers in Neuroscience*. 2024;18:1421917.

Differentiating loss of consciousness causes through artificial intelligence-enabled decoding of functional connectivity *2024*

Kim YT*, Kim H*, So M, Kong J, Kim KT, Hong JH, Son Y, Sa J, Do S, Han JH, Kim JB.

- *NeuroImage*. 2024;297:120749.

Software weakness detection in Solidity smart contracts using control and data flow analysis: A novel approach with graph neural networks *2024*

Seo A, **Kim YT**, Yang JS, Lee YS, Son Y.

- *Electronics*. 2024;13:3162.

Generating synthetic dataset for ML-based IDS using CTGAN and feature selection to protect smart IoT environments *2023*

Alabdulwahab S, **Kim YT**, Seo A, Son Y.

- *Applied Sciences*. 2023;13:10951.

SleepExpertNet: High-performance and class-balanced deep learning approach inspired by expert neurologists for sleep stage classification *2023*

Lee CH, Kim HJ, **Kim YT**, Kim H, Kim JB, Kim DJ.

- *Journal of Ambient Intelligence and Humanized Computing*. 2023;14:8067–8083.

DeepCNAP: A deep learning approach for continuous noninvasive arterial blood pressure monitoring using photoplethysmography *2022*

Kim DK, **Kim YT**, Kim H, Kim DJ.

- *IEEE Journal of Biomedical and Health Informatics*. 2022;26:3697–3707.

Intracranial densitometry-augmented machine learning enhances the prognostic value of brain CT in pediatric patients with traumatic brain injury: A retrospective pilot study *2021*

Kim YT, Kim H, Lee CH, Yoon BC, Kim JB, Choi YH, Cho WS, Oh BM, Kim DJ.

- *Frontiers in Pediatrics*. 2021;9:750272.

Prediction of life-threatening intracranial hypertension during the acute phase of traumatic brain injury using machine learning *2021*

Lee HJ, Kim H, **Kim YT**, Won K, Czosnyka M, Kim DJ.

- *IEEE Journal of Biomedical and Health Informatics*. 2021;25:3967–3976.

Cerebrospinal fluid dynamics correlates with neurogenic claudication in lumbar spinal stenosis *2021*

Kim HJ, Kim H, **Kim YT**, Sohn CH, Kim K, Kim DJ.

- *PLOS ONE*. 2021;16:e0250742.

Artifact removal from neurophysiological signals: Impact on intracranial and arterial pressure monitoring in traumatic brain injury *2020*

Lee SB, Kim H, **Kim YT**, Zeiler F, Smielewski P, Czosnyka M, Kim DJ.

- *Journal of Neurosurgery*. 2020;132:1952–1960.

Novel index for predicting mortality during the first 24 hours after traumatic brain injury *2019*

Kim H, Lee HJ, **Kim YT**, Son Y, Smielewski P, Czosnyka M, Kim DJ.

- *Journal of Neurosurgery*. 2019;131:1887–1895.

Changes in the gray and white matter of patients with ischemic-edematous insults after traumatic brain injury *2019*

Kim H, **Kim YT**, Song ES, Yoon BC, Choi YH, Kim K, Kim DJ.

- *Journal of Neurosurgery*. 2019;131:1243–1253.

Thresholds of resistance to CSF outflow in predicting shunt responsiveness *2015*

Kim DJ, Kim H, **Kim YT**, Yoon BC, Czosnyka Z, Park KW, Czosnyka M.

- *Neurological Research*. 2015;37:332–340.

Peer-Reviewed Conference Papers

BiomeTrail: Preserving Evidence, Context, and Disagreement in Mechanistic Microbe–Host Graph Question Answering *2026*

Lee J, **Kim YT**, Kim H, Kulkarni N, Skybova E, Kim V, Choi JS, Do S.

- *GenAI for Health Workshop at the Conference on Neural Information Processing Systems (NeurIPS)*. 2026. **Accepted**.

Bounded Verification and Asymmetric Recoverability in Biomedical Knowledge-Graph Construction *2026*

Lee J, **Kim YT**, Kim H, Kulkarni N, Skybova E, Kim V, Choi JS, Do S.

- *AI for Science: Verification in the Age of AI Scientists Workshop at the Conference on Neural Information Processing Systems (NeurIPS)*. 2026. **Accepted**.

Book Chapter

Functional neuromonitoring in acquired head injury *2015*

Kim H, **Kim YT**, Kim DJ.

- In: Lee S-W, Bülthoff HH, Müller K-R, editors. *Recent Progress in Brain and Cognitive Engineering*. Dordrecht, Netherlands: Springer; 2015. pp. 169–182.

Honors and Awards

People's Choice Award — Speak Your Science Competition 2025
Mass General Brigham Postdoctoral Division
- Presentation: The Missing Picture? A New Map for Microbiome Science.

Outstanding Reviewer Recognition — IEEE TIM 2025
IEEE Transactions on Instrumentation and Measurement

Minister's Award — ICT Colloquium 2020
Ministry of Science and ICT, Republic of Korea
- Automated Classification of Arrhythmia Types by Compressed ECG Representations.

Editorial and Professional Service

Editorial Service
- Topic Editor — Frontiers in Pediatrics (2025–present)
- Editorial Board Member — BMC Musculoskeletal Disorders (2025–present)

Ad Hoc Reviewer
- npj Digital Medicine
- BMC Medical Informatics and Decision Making
- BMC Medical Imaging
- Computers in Biology and Medicine
- IEEE Transactions on Instrumentation and Measurement
- Scientific Reports
- Discover Artificial Intelligence
- European Journal of Medical Research
- Scandinavian Journal of Trauma, Resuscitation and Emergency Medicine
- Neurosurgical Review
- European Spine Journal
- Discover Neuroscience
- BMC Neurology
- BMC Geriatrics
- BMC Musculoskeletal Disorders
- BMC Pulmonary Medicine
- BMC Nephrology
- BMC Pediatrics

Professional Membership
- Society for Imaging Informatics in Medicine — Member (2025–present)

Technical Program Committee
- International Conference on Electronics, Information, and Communication (ICEIC) — Member (2026)

Granted Patents

United States

Neurostimulation-based neurofeedback device using brain waves and heartbeat signals *2026*
Kim DJ, **Kim YT**, Kim HJ.
- Granted, **US 12,616,835 B2**, May 5, 2026.

Artificial intelligence neurofeedback-based telemedicine system and method of operating the same *2025*
Kim DJ, **Kim YT**, Kim HJ.
- Granted, **US 12,377,268 B2**, August 5, 2025.

Heart condition detection sensor device and system for providing complex life support solution using same *2025*
Kim DJ, **Kim YT**, Kim HJ.
- Granted, **US 12,293,833 B2**, May 6, 2025.

Republic of Korea

Neurostimulation-based neurofeedback device using brain waves and heartbeat signals *2023*
Kim DJ, **Kim YT**, Kim HJ.
- Registered, **KR 10-2556503 B1**, July 12, 2023.

Brain-computer interface apparatus and operating method of determining intention of user based on brain activity according to attention level *2023*
Kim DJ, Lee SH, **Kim YT**.
- Registered, **KR 10-2522349 B1**, April 12, 2023.

Brain-computer interface apparatus and operating method of reducing burden of individual calibration process by clustering subjects based on brain activation *2023*
Kim DJ, Lee SH, **Kim YT**.
- Registered, **KR 10-2522342 B1**, April 12, 2023.

Apparatus and method of using user-specific vagus nerve stimulation and pulsed electromagnetic field generating with AI-based neurofeedback *2023*
Kim DJ, **Kim YT**, Kim HJ.
- Registered, **KR 10-2506784 B1; KR 10-2506777 B1; KR 10-2506749 B1**, March 2, 2023. Three separate grants.

Telemedicine system and method of the same using AI-based neurofeedback *2023*
Kim DJ, **Kim YT**, Kim HJ.
- Registered, **KR 10-2506779 B1**, March 2, 2023.

Continuous blood pressure monitoring system based on photoplethysmography by using convolutional bidirectional long short-term memory neural networks *2023*
Kim DJ, Kim DK, **Kim YT**, Lee SH.
- Registered, **KR 10-2492317 B1**, January 19, 2023.

System of providing complex living support solution by monitoring and analyzing biosignal based on artificial intelligence and operating method thereof *2021*
Kim DJ, **Kim YT**, Kim HJ.
- Registered, **KR 10-2346824 B1**, December 30, 2021.

Heart condition detection sensor device and heart condition monitoring method using the same *2021*
Kim DJ, **Kim YT**, Kim HJ.
- Registered, **KR 10-2321661 B1**, October 29, 2021.

Disease prediction method and equipment using photoplethysmogram *2021*
Kim DJ, Lee SB, **Kim YT**.
- Registered, **KR 10-2296519 B1**, August 26, 2021.